

CSA15 Cloud Computing and Big Data Analytics

Capstone Cloud Technical Presentation

Individual Student Presentation Deck

Presentation Focus

Present your capstone as a cloud-backed product. Explain the problem, architecture, service choices, evidence, and CO1 learning clearly.

Presenter Details

Student Name:

Register Number:

Class / Section / Slot:

Capstone Title:

Selected SDG Goal(s):

Cloud Platform / Tools:

Guidance - delete before presentation: Fill presenter details, then delete this orange guidance tab from every slide before final submission.

Problem Context and Stakeholder Mapping

Our capstone begins with a real problem to solve

Problem Statement

What real-world problem does the product solve?

Target Users

Who will use the product and what changes for them?

SDG Fit

Which SDG goal is connected to the capstone outcome?

Product Boundary

What is included in the first usable version?

Guidance - delete before presentation: Keep the slide short. Present the problem orally with one example, not a long paragraph.

Cloud Adoption Rationale and Concept Map

The problem needs cloud support to become a usable product

Cloud Definition

Define cloud using your capstone context.

Cloud Characteristics

Elasticity, scalability, measured service, resource pooling.

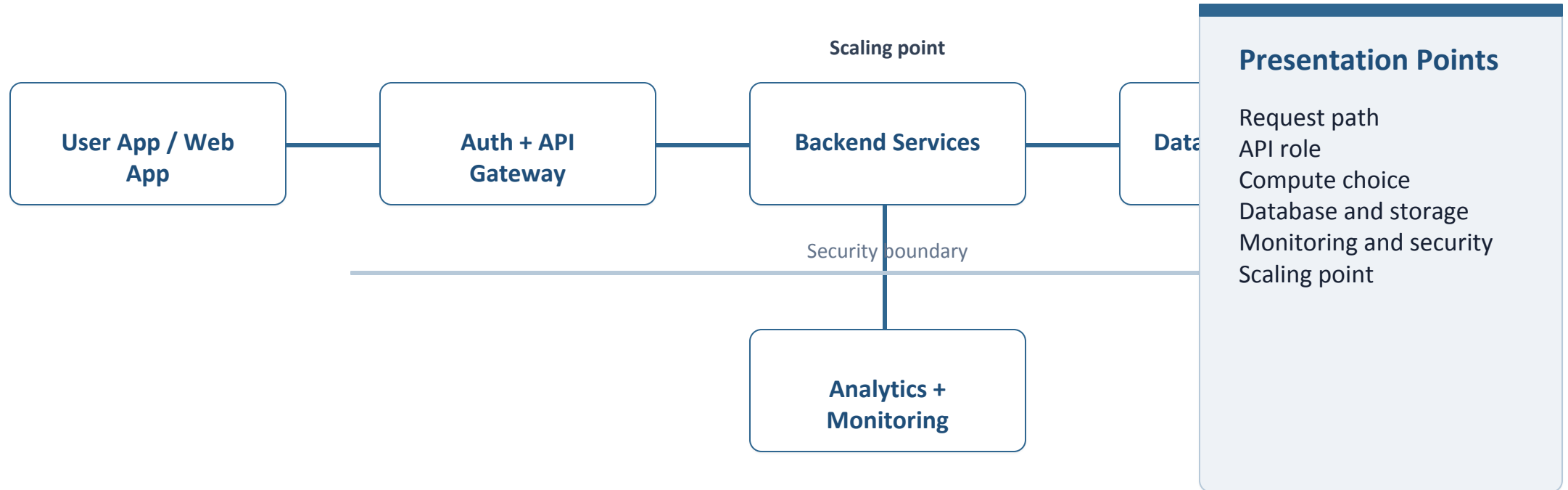
Cloud Concept Map

Place a clean concept map that links cloud ecosystem, API role, service models, elasticity, and your capstone modules.

Guidance - delete before presentation: Use the concept map as a speaking aid. Do not overload it with tiny text.

Capstone Cloud Architecture

Our cloud architecture connects the user action to services



Guidance - delete before presentation: Explain the request path from user to cloud services. Replace sample labels with your actual architecture.

Cloud Service Model Responsibility Matrix

Each service model defines what we control and what cloud manages

Service area	Model	Provider manages	Student team manages	Why this fits
Frontend hosting				
Backend/API				
Database				
Storage				
Monitoring / logs				
Security / IAM				

Guidance - delete before presentation: Mention only services that fit your capstone. Defend responsibility split orally.

Data Flow and Analytics Pipeline

The product data moves from user activity to useful insight



Data Requirements

User data, events, logs, files, sensor data, or transactions.

Analytics Requirement

Dashboard, alert, prediction, report, trend, or decision support.

Proof Required

Sample input, sample output, and one insight generated.

Guidance - delete before presentation: Show the data story. Big data analytics is optional, but the data path must be logical.

Python / AI Module Integration

Python or AI adds intelligence where the product needs it

Python Role

Prediction, automation, analytics, OCR, NLP, or reporting.

Input and Output

What data enters the module and what result comes out?

Execution Point

API endpoint, serverless function, container, notebook, or scheduled job.

Product Value

How does this module improve user experience or decision-making?

Guidance - delete before presentation: If AI is not needed, present a Python automation or analytics component.

Security, Scalability, Privacy and Cost Design

Security, scale, privacy and cost shape the final design

Security

Authentication, roles, validation, audit trail.

Privacy

Sensitive data, consent, anonymization, access limits.

Scalability

How will users grow without redesign?

Cost Awareness

Free tier, managed services, storage/API usage limits.

Guidance - delete before presentation: Present decisions, not generic definitions. Tie every point to the capstone.

Implementation Stack and Deployment Evidence

Our implementation stack shows how the design will be built

Mobile / Web Layer

Android, iOS, Flutter, React, or web frontend.

Cloud Services

Compute, database, storage, authentication, analytics, API tools.

Evidence

GitHub, API test, cloud dashboard, endpoint, logs, or deployment URL.

Version Control

Repository structure, commits, branches, and README clarity.

Guidance - delete before presentation: Show real evidence. One clear screenshot is better than many unreadable screenshots.

Demo Storyboard and User Journey

A complete user journey proves the product flow

Scene 1

User action -> system response -> cloud component touched

Scene 2

User action -> system response -> cloud component touched

Scene 3

User action -> system response -> cloud component touched

Scene 4

User action -> system response -> cloud component touched

Scene 5

User action -> system response -> cloud component touched

Scene 6

User action -> system response -> cloud component touched

Guidance - delete before presentation: Show the product journey. Each scene should connect user action, system response, and cloud component.

Evaluation Readiness Checklist

Our evidence confirms that the presentation is ready

☐

Cloud ecosystem and service model explained

☐

Elasticity and scalability decisions are visible

☐

REST/SOAP/API role connected to capstone

☐

GitHub and cloud evidence are inspectable

☐

Architecture has compute, data, storage, security, monitoring

☐

References and integrity statement are included

Evaluator Focus

CO1 is verified through cloud design reasoning, service choices, evidence, and individual explanation.

Guidance - delete before presentation: Use this as a self-check slide while preparing. Delete the guidance tab before presentation.

Conclusion, Limitations and Future Scope

We close with learning, limitations and next steps

Learning Outcome

What cloud concepts did the project make clear?

Limitations

What remains incomplete or needs improvement?

Future Improvement

What will be improved before final capstone review?

Integrity Statement

Mention sources, tools, datasets, and any AI assistance acknowledged.

Guidance - delete before presentation: End with technical learning, not only a thank-you slide.